

ITE 3822 - Discrete Mathematics
Assignment 2

- Answer All Questions.
 - Only **HANDWRITTEN** answers will be accepted.
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1. Consider the matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}.$$

- (i) Find the determinant of the matrix $\mathbf{P}$.
 - (ii) Find $\text{adj}(\mathbf{P})$.
 - (iii) Hence find $\mathbf{P}^{-1}$.
2. There are ten students A to J in an undergraduate mathematics program. At the end of the semester, the students are to take exams for the subjects they are enrolled in. A , B and I took Probability; A , D , I and J took Graph Theory; C , D , F and H took Henstock Integration; E and F took Lie Groups; B , E and G took Non-linear Dynamical Systems; D and J took Topology. The exam must be scheduled in such a way that no student has to take two subjects at the same time. Use the concepts of graphs to answer the following questions.
- (a) What will the vertices represent?
 - (b) Under what condition will two vertices be joined by an edge?
 - (c) Construct a graph model for the problem.
3. Prove that the relation R defined on $\mathbb{Z}$ given by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation.